

```

//importing needed libs
import java.io.File;
import java.io.FileNotFoundException;
import java.io.PrintWriter;
import java.util.ArrayList;
import java.util.Scanner;

//creating my class
public class Zoo {
    Scanner sc = new Scanner(System.in);
    //creating scanner object
    //creating class variables...
    public ArrayList<Animal> animals = new ArrayList<>(); //creating my array list

    private String address = "Unknown";
    private String telephone = "Unknown";
    private String zooName = "Ulster Zoo"; // default
    private String postcode = "Unknown";

    //loading animals from a file
    public void loadAnimals() {
        animals.clear();
        try {
            File file = new File("AnimalDetails.txt");
            if (!file.exists()) {
                System.out.println("AnimalDetails.txt not found.");
                return;
            } //end of if
            Scanner reader = new Scanner(file);
            //reading the file
            int lineNumber = 0;
            while (reader.hasNextLine()) {
                //creating a thing to hold the string line to sort through
                String line = reader.nextLine();
                lineNumber++;
                if (line == null || line.strip().isEmpty()) continue;
                String[] p = line.split(";");
                //splitting the string line into words to sort through
                for(int i = 0; i < p.length;i++){
                    p[i] = p[i].trim();
                } //end of for
            }
        }
    }
}

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String name = p[0]; //the name of the animal is at location 0
int age = Integer.parseInt(p[1]); //same for age
String color = p[2]; //same for color
double weight = Double.parseDouble(p[3]); //same for weight
String type = p[4]; //same for type

//applying special vars to the locations
if (type.equalsIgnoreCase("Tiger")) {
    int clawLength = Integer.parseInt(p[5]); //location 5, special var
    animals.add(new Tiger(name, age, color, weight, clawLength));
} //end of if
else if (type.equalsIgnoreCase("Parrot")) {
    String featherCondition = p[5]; //same thing
    animals.add(new Parrot(name, age, color, weight, featherCondition));
} //end of else if
else if (type.equalsIgnoreCase("Shark")) {
    int teethHealth = Integer.parseInt(p[5]); //same again
    animals.add(new Shark(name, age, color, weight, teethHealth));
} //end of final else if
} //end of while
//closing the file reader.
reader.close();

//catching the exceptions
} //end of try
catch (Exception e) {
    System.out.println("No previous animal file found.");
    //catching the exception if animal is not found.
} //end of catch
} //end of load animals

//saving the animals to the file
public void saveAnimals() {

    if (animals == null || animals.isEmpty()) {
        System.out.println("No animals to save. File not overwritten.");
        return;
    } //end of checking for nulls or empty

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try (PrintWriter pw = new PrintWriter("AnimalDetails.txt")) {
    for (Animal a : animals) {
        if (a.isValid()){
            pw.println(a);
        }
        //I am allowing users to add numbers to animal names in case of the event that
        //animals have the same name, such as Jack1, Jack2
        else{
            System.out.println("The animal can not have null values " +
                "or the value can not equal 0.");
        }
    }
} //end of for
} //end of try
catch (Exception e) {
    System.out.println("Error saving animals.");
} //end of catch
} //end of saveAnimals

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public void addAnimal(Animal a) {
    if (a == null){
        System.out.print("Can not save animals, as it contains null values.");
    } //end of check for nulls
    else if (!a.isValid()){
        System.out.println("Animal contains invalid values.");
        return;
    } //end of check for valid.
    else if (a.isValid()){
        System.out.println("Animal values correct, animal saved successfully.");
        animals.add(a);
    } //end of check for valid
    else{
        System.out.print("An error has occurred");
        //Don't think this would ever run, added just in case
    } //end of final check
    //adding the animal
} //end of add animals

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public void removeAnimal(String name) {
    boolean found1 = false;

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for (int i = 0; i < animals.size(); i++) {
    if (animals.get(i).getName().equalsIgnoreCase(name)) {
        //adding ignore case to stop error...
        animals.remove(i);
        found1 = true;
        break;
    } //end of if
} //end of for
if (found1 == false){
    System.out.println("Animal can not be found.");
} //end of if
else if (found1 == true){
    System.out.println("Animal has been removed");
} // end of else if
} //end of remove animals

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public Animal searchByName(String name) {
    for (int i = 0; i < animals.size(); i++) { //sorting by length of arraylist
        Animal a = animals.get(i);
        //attaching animal to a variable
        if (a.getName().equalsIgnoreCase(name)) {
            return a; //if its found
        } //end of if
    } //end of for
    System.out.println("Animal not found or null values inputted.");
    return null;
    //if not found
} //end of search by name

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public void dailyCare() {
    for (Animal a : animals) {
        System.out.println("General care for " + a.getName());
        if (a instanceof Flyable f) f.checkWings();
        if (a instanceof Swimmable s) s.checkFins();
    } //end of for
} //end of daily care

```

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public void report() {

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//print the hardcoded zoo name
System.out.println("Zoo Report for " + zooName);

//creating a list for the counts and types
ArrayList<String> types = new ArrayList<String>();
ArrayList<Integer> counts = new ArrayList<Integer>();

//sorting through animals to find the animal name
for (int i = 0; i < animals.size(); i++) {
    String type = animals.get(i).getClass().getSimpleName();

    //creating an index as it should show no animal being found

    int index = -1;

    //looks for the animal in the list
    for (int j = 0; j < types.size(); j++) {
        if (types.get(j).equals(type)) {
            index = j;
            break;
        } //end of if
    } //end of for

    if (index == -1) {
        //if the animal is not found in the index, it adds it.
        types.add(type);
        counts.add(1);
    } //end if statement
    else {
        //if the animal already exists, it increases the count
        counts.set(index, counts.get(index) + 1);
    } //end of else
} //end of for

//prints the count for each type of animal.
for (int i = 0; i < types.size(); i++) {
    System.out.println(types.get(i) + " the count is " + counts.get(i));
} //end of for

//find the dominating color
//setting the string to none, in case of two equal amounts ect

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String dominantColor = "None";
//setting the dominant count
int dominantCount = 0;
//sorting through animals to et each color for the length of animals
for (int i = 0; i < animals.size(); i++) {
    String color = animals.get(i).getColor();
    int colorCount = 0;
    //incrementing the colors for each count of animal.
    for (int j = 0; j < animals.size(); j++) {
        if (animals.get(j).getColor().equalsIgnoreCase(color)) {
            colorCount++;
        } //end of if
    } //end of for

    if (colorCount > dominantCount) {
        dominantCount = colorCount;
        dominantColor = color;
        //setting the dominant color and count so it can be printed
    } //end of if
} //end of for

System.out.println("Dominant color: " + dominantColor);
//printing the dominant color...
} //end of report

public void listAnimals() {
    for (int i = 0; i < animals.size(); i++) {
        Animal a = animals.get(i);
        System.out.println(a.makeSound());
    } //end of for
} //end of list animals

public void loadZooDetails(){
    try (Scanner reader = new Scanner(new File("ZooDetails.txt"))){
        while (reader.hasNextLine()){
            String line = reader.nextLine().trim();
            if (line.isEmpty()) continue;
            String[] parts = line.split(";");
            if (parts.length >= 4) {
                this.address = parts[0].trim();
                this.telephone = parts[1].trim();
            }
        }
    }
}

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        this.zooName = parts[2].trim();
        this.postcode = parts[3].trim();
    } //end of if
} //end of while,
} //end of try
catch (Exception e){
    System.out.println("ZooDetails.txt can not be found");
}
} //end of function

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```

public ArrayList<Animal> searchByColor(String color) {
    ArrayList<Animal> results = new ArrayList<Animal>();
    for (int i = 0; i < animals.size(); i++) {
        Animal a = animals.get(i);
        if (a.getColor().equalsIgnoreCase(color)) {
            results.add(a);
        } //end of if
    } //end of for
    if (results.isEmpty()){
        System.out.println("Animal colour not found or null values inputted.");
    }
    return results;
} //searching by color by checking if animal color is matching
//qual to the color selected

```

```

public void saveZooDetails() {
    try(PrintWriter pw = new PrintWriter("ZooDetails.txt")) {
        pw.println(address + "," + telephone + "," + zooName + "," + postcode);
    } catch (Exception e) {
        System.out.println("Unable to save ZooDetails.txt");
    } //end of exception
} // end of function

```

```

public void AnimalMovement(){
    for (Animal a : animals) {
        if (a instanceof Swimmable s) s.swim();
        else if (a instanceof Flyable f) f.fly();
        else{
            System.out.println(a.getName() + " gallops gracefully.");
        }
    }
}

```

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        }//end of else

    }//end of for
} //end of animal movement

public void printZooDetails() {
    try {
        File file = new File("ZooDetails.txt");
        if (!file.exists()) {
            System.out.println("ZooDetails.txt not found.");
            return;
        } //end of if
        Scanner reader = new Scanner(file);

        while (reader.hasNextLine()){
            String line = reader.nextLine();
            String[] str = line.split(",");
            String address = str[0];
            System.out.println("Address: " + address);

            String phone = str[1];
            System.out.println("Phone: " + phone);

            String name = str[2];
            System.out.println("Name of Zoo: " + name);

            String postCode = str[3];
            System.out.println("Postcode: " + postCode);
        } //end of while
    } //end of try
    catch (FileNotFoundException e){
        System.out.println("The file can not be found");
    } //end of catch

} //end of print zoo details

public void getNewZooDetails(){
    System.out.println("
    What would you like to modify?
    1. Address

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2. Telephone

3. Zoo Name

4. Postcode

0. Exit

""");

```
int choicei = sc.nextInt(); //getting choice
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```
if (choicei == 0){
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    System.out.println("Stopping the Update of Zoo details");
```

```
    return;
```

```
}//end of if
```

```
if (choicei == 1) {
```

```
    sc.nextLine();
```

```
    System.out.println("Please Input the new Address: ");
```

```
    String newAddress = sc.nextLine().trim();
```

```
    if (!newAddress.isEmpty()) {
```

```
        this.address = newAddress;
```

```
        System.out.println("Address Updates");
```

```
    } //end of check for null
```

```
    else {
```

```
        System.out.println("The Zoo Details will not be updated as " +
```

```
            "either null values, empty values " +
```

```
            "or incorrect values have been inputted.");
```

```
    } //end of else
```

```
} //end of if 1
```

```
if (choicei == 2) {
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```
    sc.nextLine();
```

```
    System.out.println("New Telephone: ");
```

```
    String newTelephone = sc.nextLine().trim();
```

```
    if (!newTelephone.isEmpty()) {
```

```
        this.telephone = newTelephone;
```

```
        System.out.println("Telephone Updated");
```

```
    } //end of check for null
```

```
    else {
```

```
        System.out.println("The Zoo Details will not be updated as " +
```

```
            "either null values, empty values " +
```

```
            "or incorrect values have been inputted.");
```

```
    } //end of else
```

```
} //end of if 2
```

```
if (choicei == 3) {
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```
    sc.nextLine();
```

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        System.out.println("Please Input the new Zoo Name: ");
        String newZooName = sc.nextLine();
        if (!newZooName.isEmpty()){
            this.zooName = newZooName;
            System.out.println("Zoos name Updated");
        } //end of check for null
        else {
            System.out.println("The Zoo Details will not be updated as " +
                "either null values, empty values " +
                "or incorrect values have been inputted.");
        } //end of else
    } //end of if 3
    if (choicei == 4) {
        sc.nextLine();
        System.out.println("Please Input the new Postcode: ");
        String newPostcode = sc.nextLine();
        if (!newPostcode.isEmpty()) {
            this.postcode = newPostcode;
            System.out.println("Postcode Updated");

        } //end of check for null
        else {
            System.out.println("The Zoo Details will not be updated as " +
                "either null values, empty values " +
                "or incorrect values have been inputted.");
        } //end of else
    } //end of if 4
} //end of get new zoo details
//setting functions for testing zoo details
public String getZooName () {
    return zooName;
    //using this to test to see if zoo name is null during testing
} //end of returning zooName
public String getAddress () {
    return address;
    //using this to test to see if zoo address is null during testing
} //emd of get address
public String getTelephone () {
    return telephone;
    //attaching telephone to the get Telephone so it can be tested for null values in
testing

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```
    } //end of get telephone
    public String getPostcode () {
        return postcode;
        //attaching get post code to the variable getPostcode so it can be tested for nulls
        //in testing
    } //end of getPostcode testing method
    //creating setts for get zoo details for testing purposes.
    public void setZooName(String zooName) {
        this.zooName = zooName;
    } //setting the zoos name
    public void setAddress(String address) {
        this.address = address;
    } //setting the zoos address
    public void setTelephone(String telephone) {
        this.telephone = telephone;
    } //setting the zoos telephone
    public void setPostcode(String postcode) {
        this.postcode = postcode;
    } //setting the zoos postcode
} //end of class
```